

Q1

[org 0x0100]

jmp start

printing:

push es

push ax

push cx

push di

mov ax, 0xb800

mov es, ax

xor di, di

mov ah, 0x07

mov al, '_'

mov cx, 1120

print:

stosw

loop print

mov ah, 0x07

mov al, '.'

mov cx, 2000

print2:

stosw

loop print2

pop di

pop cx

pop ax

pop es

ret

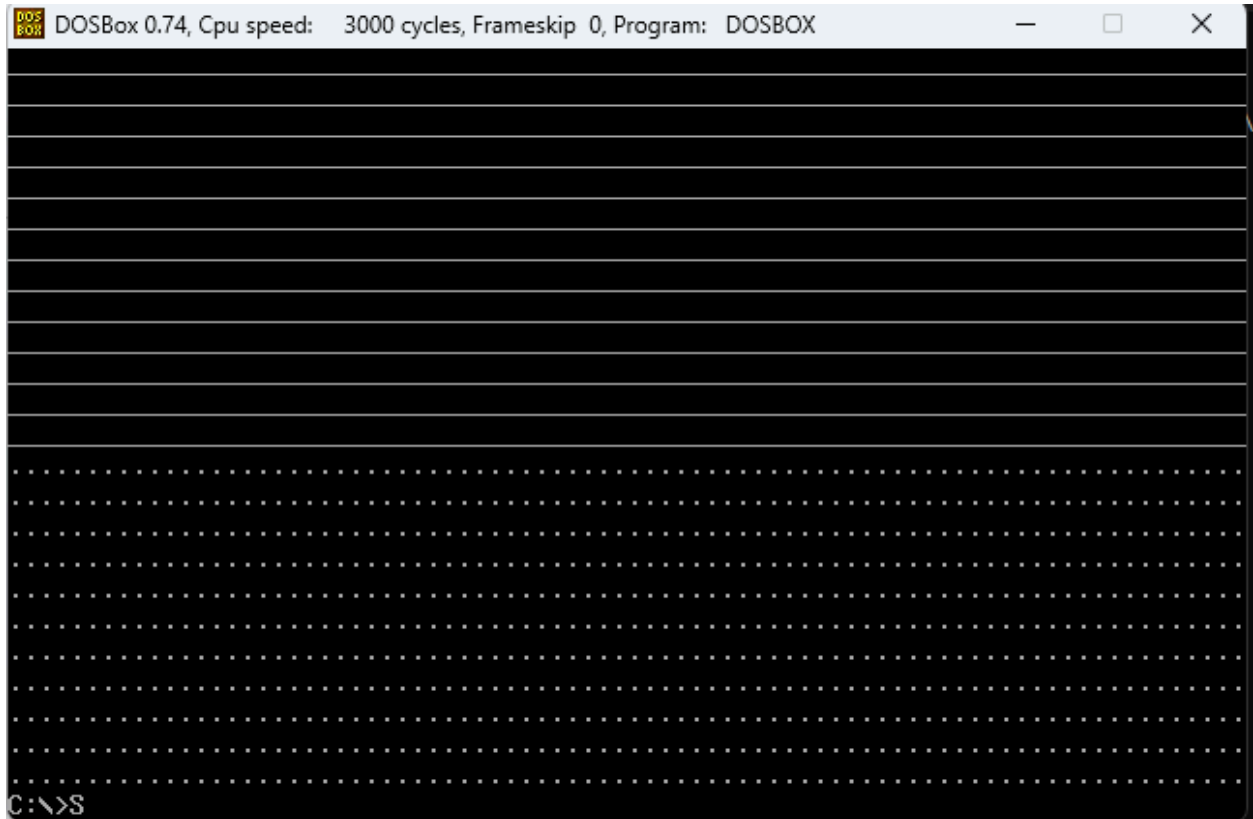
start:

call clrscr

end:

mov ax, 0x4c00

int 0x21



Q2

```
[org 0x0100]  
jmp start
```

```
delay:
```

```
push bp
```

```
mov bp, sp
```

```
push ax
```

```
push bx
```

```
push cx
```

```
mov bx, [bp+4]
```

```
outer_loop:
```

```
mov cx, 65535
```

```
inner_loop:
```

```
loop inner_loop
```

```
dec bx
```

```
cmp bx, 0
```

```
jnz outer_loop
```

```
pop cx
```

```
pop bx
```

```
pop ax
```

```
pop bp
```

```
ret 2
```

scrollup:

```
push bp
mov bp, sp
push ax
push cx
push si
push di
push es
push ds
```

```
mov ax, 80
mul byte [bp+4]
mov si, ax
push si
shl si, 1
mov cx, 2000
sub cx, ax
mov ax, 0xb800
mov es, ax
mov ds, ax
xor di, di
cld
rep movsw
mov ax, 0x0720
pop cx
rep stosw
```

```
pop ds
pop es
pop di
pop si
pop cx
pop ax
pop bp
ret 2
```

scrolldown:

```
push bp
mov bp, sp
push ax
push cx
push si
```

```
push di
push es
push ds
```

```
mov ax, 80
mul byte [bp+4]
push ax
shl ax, 1
mov si, 3998
sub si, ax
mov cx, 2000
sub cx, ax
mov ax, 0xb800
mov es, ax
mov ds, ax
mov di, 3998
std
rep movsw
mov ax, 0x0720
pop cx
rep stosw
```

```
cld
pop ds
pop es
pop di
pop si
pop cx
pop ax
pop bp
ret 2
```

start:

looping:

```
mov ax, 50
push ax
call delay
push byte 1
call scroll down
mov ax, 50
push ax
call delay
```

push byte 1
call scrollup

jmp looping